

NAN AGROMET, Thailand

WMO: 483330

Lat: 18.8641N

Lon: 100.7416E

Elev: 870

StdP: 14.24

Time Zone: 7.00 (E07)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	53.2	56.4	48.6	52.3	62.1	51.9	59.3	64.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	17.3	99.3	76.6	97.1	76.3	95.0	76.6	81.0	90.7	80.4	89.8	79.9	88.9	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.8	154.4	86.1	78.1	151.0	85.0	77.5	147.9	84.1	45.7	90.7	44.9	90.2	44.3	89.4	88.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
N/A	N/A	N/A	DB	46.9	101.7	3.6	2.9	44.3	103.8	42.3	105.5	40.3	107.1	37.7	109.2	
			WB	46.7	83.3	3.9	1.9	43.9	84.6	41.6	85.7	39.5	86.8	36.7	88.1	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	79.4	71.4	75.3	79.7	83.8	84.0	83.2	81.7	81.3	81.8	80.7	77.3	72.2	(6)
	DBStd	5.23	4.24	3.37	3.86	3.24	2.98	2.51	2.50	2.02	1.83	2.21	3.57	4.31	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	15	8	0	1	0	0	0	0	0	0	0	0	6	(9)
	CDD50	10721	664	708	920	1013	1053	996	984	970	955	951	818	689	(10)
	CDD65	5260	207	288	456	563	588	546	519	505	505	486	368	229	(11)
	CDH74	53389	1675	2896	5071	6560	6568	6201	5314	4851	4957	4471	3118	1707	(12)
	CDH80	22031	506	1308	2718	3559	3083	2508	1791	1485	1689	1674	1198	512	(13)
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(14)
(15) Precipitation	PrecAvg	51.9	0.5	0.4	1.7	3.8	7.1	5.7	8.9	11.3	8.5	2.7	0.9	0.5	(15)
	PrecMax	76.1	2.4	2.9	6.1	9.0	15.2	10.9	15.6	22.0	15.5	5.0	5.9	3.7	(16)
	PrecMin	39.9	0.0	0.0	0.0	0.9	2.6	2.3	3.7	5.2	4.8	0.3	0.0	0.0	(17)
	PrecStd	8.1	0.7	0.6	1.5	1.9	3.1	2.2	3.0	4.0	2.6	1.1	1.2	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	89.3	94.9	99.7	103.0	101.5	97.4	95.7	92.2	92.9	93.2	92.5	89.6	(19)
		MCWB	71.2	71.9	72.8	77.1	79.0	79.5	78.4	79.7	79.5	78.4	76.7	73.2	(20)
	2%	DB	87.2	92.7	97.2	100.2	98.0	94.4	92.6	90.1	91.0	91.4	90.4	87.4	(21)
		MCWB	70.5	71.8	73.2	76.9	78.5	79.4	78.9	79.4	79.3	78.1	76.2	72.8	(22)
	5%	DB	85.3	90.6	95.5	97.9	95.1	92.2	90.0	88.5	89.6	90.0	88.9	85.4	(23)
		MCWB	69.8	71.5	73.4	76.9	78.6	79.2	78.9	79.0	79.1	77.8	75.6	71.9	(24)
	10%	DB	83.1	88.1	93.0	95.1	92.4	90.0	87.8	86.9	88.0	88.3	87.0	83.1	(25)
		MCWB	69.3	70.8	73.2	76.9	78.5	78.9	78.6	78.7	78.7	77.4	74.8	70.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.9	76.1	78.4	81.7	81.8	81.7	81.0	81.4	81.4	80.8	78.8	77.2	(27)
		MCDB	84.7	89.4	92.5	94.8	93.3	92.1	89.8	88.8	89.4	88.9	88.9	85.6	(28)
	2%	WB	72.8	74.7	76.7	79.9	80.6	80.7	80.1	80.3	80.4	79.2	77.8	75.3	(29)
		MCDB	82.5	87.6	91.4	93.8	92.2	90.3	88.3	87.5	88.3	87.8	87.3	83.2	(30)
	5%	WB	71.7	73.4	75.6	78.9	79.7	79.9	79.4	79.6	79.7	78.6	77.0	73.6	(31)
		MCDB	80.8	86.0	90.1	92.7	91.0	89.0	87.2	86.5	87.2	87.0	86.1	81.6	(32)
	10%	WB	70.6	72.1	74.5	77.8	78.9	79.2	78.8	79.0	79.0	78.0	76.1	72.2	(33)
		MCDB	79.5	84.5	88.4	91.2	89.9	87.7	86.2	85.6	86.2	85.7	84.4	80.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	24.5	28.2	27.4	23.0	17.3	13.5	11.7	11.5	13.2	16.2	20.6	22.9	(35)
		MCDBR	27.4	31.4	31.6	27.5	22.1	17.3	15.8	14.2	15.6	18.6	22.0	24.4	(36)
	5% WB	MCWBR	11.5	12.1	10.4	7.9	5.9	4.7	4.8	5.1	5.3	6.8	8.8	10.7	(37)
		MCWBR	22.5	27.3	27.2	24.0	18.9	15.2	13.6	13.0	14.5	16.6	19.6	20.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.438	0.537	0.685	0.653	0.491	0.442	0.437	0.454	0.481	0.518	0.425	0.424	(40)
		taud	2.200	1.936	1.623	1.744	2.212	2.378	2.383	2.317	2.225	2.079	2.301	2.263	(41)
	Ebn at Noon		265	244	214	223	259	270	271	268	261	246	268	266	(42)
		Edh at Noon	43	59	84	75	46	39	39	42	45	51	39	40	(43)
(44) All-Sky Solar Radiation	RadAvg		1411	1626	1691	1836	1676	1504	1383	1369	1447	1454	1464	1348	(44)
	RadStd		82	79	100	122	152	115	113	89	68	72	88	79	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	+0.56	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+265	+268

Nomenclature: See separate page